6.2.2 Stochastic Integration

In order to develop good sampling paths for the Gibbs-Boltzmann distribution, we need to introduce a method of adding random perturbations into a differential equation. Let us first consider the definition of the integral *with respect to Wiener measure*.

Recall that the Riemann integral of a continuous function g on $[0, \tau]$ is the limit of Riemann sums constructed as follows: partition the interval $[0, \tau]$ into K subintervals each of width $\delta t = \tau/\nu$, choose an arbitrary t_k from the kth subinterval, $1 \leq k \leq \nu$, and evaluate the integral as a limit of the sum of products:

$$\int_0^\tau g(t)\mathrm{d}t = \lim_{\nu\to\infty} \sum_{k=0}^{\nu-1} g(t_k)\delta t.$$

We know that this limit does not depend on where the t_k are chosen within the subintervals. For example we can take the left or right endpoint of each subinterval or the midpoint.

By analogy, we would like to write the stochastic integral of a function g, where g may, itself, be defined in terms of a stochastic process, as the limit of a sum:

$$\sum_{k=0}^{\nu-1} g(t_k)[\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t)], \qquad \nu \to \infty,$$

where the increments $\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t)$ identified here are the increments of a Wiener process whose properties are given above. (In particular they are normally distributed with mean zero and variance δt.) First, we should be careful regarding the sense in which we expect convergence, since sequences of random variables may converge in various ways.

Next, unlike in the case of the Riemann sums, it turns out that the result depends on where we place the points t_k. Using the properties of the Wiener process we may easily show that the choice $t_k = k\delta t$ (left endpoint) and the choice $t_k = (k + 1/2)\delta t$ yield different formulas for the integral. The first choice is commonly referred to as *Itō integration*, and the second as *Stratonovich integration*. As an example, consider the case where $g(t) = \mathrm{W}(t)$. We suppose this to be approximated by a sum

$$\begin{aligned}\int_0^\tau \mathrm{W}(t)\mathrm{dW}(t) &\approx \sum_{k=0}^{\nu-1} \mathrm{W}(k\delta t)[\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t)], \\ &= \frac{1}{2}\sum_{k=0}^{\nu-1}\Big[\big(\mathrm{W}((k+1)\delta t)^2 - \mathrm{W}(k\delta t)^2\big) - \big(\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t)\big)^2\Big].\end{aligned}$$

Breaking this into two parts, we see that the first sum telescopes to give $\mathrm{W}(\tau)^2$ (as $\mathrm{W}(0) = 0$). On the other hand, from (6.7), we have

$$\lim_{\nu\to\infty}\mathbb{E}\sum_{k=0}^{\nu-1}\Big[(\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t))^2\Big] = \tau,$$

but an even stronger statement is possible. Since we are summing a large collection of random variables $\Delta\mathrm{W}_k$ whose means are δt, the Central Limit Theorem tells us that, if we assume it to be well defined, the variance of the sum of squares from τ must tend to zero as a function of ν. Thus we may understand the convergence in a somewhat stronger sense than convergence in expectation. We make this precise in the following proposition

Proposition 6.2 *Let $\nu\delta t = \tau$ then*

$$\text{l.i.m.}_{\nu\to\infty}\sum_{k=0}^{\nu-1}\Big[(\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t))^2\Big] = \tau,$$

where, for a sequence of random variables, A_ν *and a random variable* B*, we have*

$$\text{l.i.m.}_{\nu\to\infty}\mathrm{A}_\nu = \mathrm{B} \Leftrightarrow \lim_{\nu\to\infty}\mathbb{E}(\mathrm{A}_\nu - \mathrm{B})^2 = 0.$$

(This is referred to as convergence in the mean square sense.)

Proof With τ, ν fixed, we have $\delta t = \tau/\nu$.

Let $\Delta\mathrm{W}_{\nu,k} = \mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t) = \mathrm{W}((k+1)\tau/\nu) - \mathrm{W}(k\tau/\nu)$ and observe that

$$\sum_{k=0}^{\nu-1}\Delta\mathrm{W}_{\nu,k}^2 - \tau = \sum_{k=0}^{\nu-1}\Big[\Delta\mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu}\Big].$$

Then

$$I_\nu \overset{\text{def}}{=} \mathbb{E}\left(\sum_{k=0}^{\nu-1}\Delta\mathrm{W}_{\nu,k}^2 - \tau\right)^2 = \mathbb{E}\left(\sum_{k=0}^{\nu-1}\Big[\Delta\mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu}\Big]\right)^2.$$

Terms of the form $Z_{\nu,k} \overset{\text{def}}{=} \Delta\mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu}$ are independent and have mean zero. Therefore

$$\mathbb{E}\left(\sum_{k=0}^{\nu-1}Z_{\nu,k}\right)^2 = \mathbb{E}\sum_{k=0}^{\nu-1}Z_{\nu,k}^2 - 2\mathbb{E}\sum_{k\neq l}Z_{\nu,k}Z_{\nu,l} = \mathbb{E}\sum_{k=0}^{\nu-1}Z_{\nu,k}^2,$$